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Research article

An event-triggered deep reinforcement learning guidance algorithm for intercepting maneuvering target of speed advantage

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ABSTRACT

Hypersonic vehicle interception raises stringent and challenging requirements for traditional guidance laws in terms of speed and maneuverability advantage. Deep reinforcement learning algorithms provide potential solutions for intercepting maneuvering targets of speed advantage, but they are greatly hindered by policy training inefficiency. To address these limitations, we propose a novel event-triggered deep reinforcement learning (ETDRL) algorithm along with an event-triggered training and time-triggered execution (ETTE) framework. The ETTE framework reformulates the agent-environment interaction as an event-triggered Markov decision process (ETMDP) model, where the agent updates its action only when the environment state meets a specific event triggering condition, otherwise maintaining the previous behavior between events. As a result, this approach significantly accelerates policy training by reducing the total number of decision steps required during the learning phase. To mitigate the potential degradation in control performance caused by the event-triggered mechanism, the ETTE framework enables well-trained policies to be executed with a fixed decision interval, that is, in a time-triggered way. Based on the proposed method, an ETDRL guidance law is developed for intercepting maneuvering targets of speed advantage under constraints of limited maneuverability, large initial heading error, and bearings-only measurement. By following the design principle of nullifying the line-of-sight angular rate to establish a collision course with the target, we model the guidance problem as an ETMDP. The twin delayed deep deterministic policy gradient algorithm is utilized to train the ETDRL guidance law. Numerical simulations demonstrate the superiority of the proposed ETDRL method over DRL algorithms in terms of policy training efficiency, while also highlighting its enhanced guidance performance over traditional methods.

1. Introduction

The recent advancements in hypersonic vehicles have introduced significant challenges for missile defense systems, particularly in the realm of intercept guidance. The hypersonic vehicles' high speed and maneuverability make it difficult for existing guidance laws to achieve precise interception. Specifically, since the interception engagement scenario is usually head-on, the high-speed nature of these targets results in a rapid closing velocity and shortened engagement time, placing increased demands on the missile's guidance system [1]. Furthermore, the missile must execute necessary maneuvers to counteract the target maneuver, while it often faces limited maneuverability in endoatmospheric engagements. This disparity highlights the inherent limitations of existing guidance laws, which are primarily designed under the assumption that the missile exhibits both speed and maneuverability advantages over the target. Once this condition is violated, their performance degrades dramatically. As a result, it is of practical significance to develop a guidance law tailored for intercepting targets of speed advantage to enhance the capturability of missiles against hypersonic vehicles.

As the most widely used guidance law, proportional navigation (PN) [2] has attracted much attention from researchers and has inspired the development of various variants of PN [3–5] primarily due to its simplicity of implementation. To address the challenge of capturing targets with uncertain maneuvers, augmented proportional navigation (APN) was developed in [6,7] through the incorporation of a compensation term for target acceleration. In [8], the capturability of augmented pure proportional navigation (APPN) in head-on engagements against maneuvering targets of speed advantage was analyzed, revealing that its capture region becomes significantly reduced as target

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velocity increases. Furthermore, to enhance the robustness to the uncertain target maneuver, nonlinear control theory has been leveraged to develop guidance laws, such as H_∞ control [9,10] and sliding mode control [11–19]. These methods typically treat unknown target acceleration as a bounded external disturbance, necessitating its upper bound as prior knowledge for guidance law design. However, since the upper bound of target acceleration is often unavailable in practice, researchers have explored composite control methods that incorporate adaptive control [14,15,16] or extended disturbance observer [17,18,19], where the target maneuver are estimated adaptively and then fed back to the guidance command. Despite these advancements, the hypersonic vehicles of complex and changeable trajectories bring challenges to the estimation of target maneuvers in both computation complexity and accuracy. In addition, the aforementioned methods rely on an ideal assumption that missiles show a substantial acceleration advantage over targets. This condition is difficult to guarantee in real-world scenarios, and it will lead to a large miss distance once the condition is violated. In summary, the intercept guidance problem against maneuvering targets of speed advantage with limited maneuverability remains unresolved.

Reinforcement learning (RL) [20] is an effective approach to addressing decision-making and control problems, and it offers a novel methodology for solving the guidance problem. RL algorithms learn an optimal or suboptimal strategy from scratch through interactions with unknown environments. By integrating deep neural networks, deep reinforcement learning (DRL) algorithms, such as deep Q-network (DQN) [21], deep deterministic policy gradient (DDPG) [22], twin delayed DDPG (TD3) [23] proximal policy optimization (PPO) [24] and hierarchical reinforcement learning (HRL) [25,26], are of powerful representation ability to map raw sensory inputs directly to policies without additional information processing. By introducing DRL algorithms into the guidance law design, numerous works have demonstrated that DRL guidance laws are capable of solving the intercept guidance problem against maneuvering targets and outperform traditional methods in both hit probability and interception accuracy [27–35]. In [27–32], guidance laws for intercepting maneuvering targets were developed by using various DRL approaches. To further improve the lethality of missiles, the impact time and angle control guidance laws were proposed in [33–35]. These findings demonstrate that a distinctive feature of DRL guidance laws is their ability to operate without requiring any prior knowledge about the system model, especially the target maneuver model. Instead, they leverage neural networks' representation capabilities to extract high-level useful information from raw sensory data, enabling an end-to-end mapping from the seeker's measurement to the guidance command. Given these advantages, DRL algorithms have been employed to investigate guidance laws against hypersonic vehicles, where targets typically exhibit a speed advantage over interceptors. In [36] and [37], the PPO algorithm was utilized to optimize the guidance policy under a two-dimensional head-on engagement scenario. In [38], a three-dimensional PPO-based guidance law for bearings-only interception was proposed, directly mapping the line-of-sight (LOS) angle and its rate obtained from an onboard passive infrared seeker to the missile's commanded thrust. This approach demonstrated superior performance compared to the augmented zero-effort miss guidance with perfect target acceleration knowledge. However, these existing methods require substantial interceptor maneuverability advantages to address issues stemming from targets' high speed and large initial heading errors.

To address the issues of limited maneuverability, this work aims to solve the three-dimensional intercept guidance problem against maneuvering targets of speed advantage by using DRL algorithms. However, existing DRL algorithms are greatly hindered by learning inefficiency when tackling this complex guidance problem. On one hand, the three-dimensional engagement scenario results in an excessively large state space and action space, leading to low exploration efficiency. On the other hand, constraints such as speed disadvantage, limited maneuverability, large heading error, and bearings-only measurement, significantly narrow the feasible solution domain. These issues cause the learning process to be prolonged and inefficient, adversely affecting the interception performance of DRL guidance laws. Moreover, traditional DRL algorithms are typically based on the Markov decision process (MDP) model [39]. In this framework, the agent interacts with the environment in a time-triggered manner, that is, making decisions at fixed intervals and focusing solely on the current state. For high-dynamic continuous control problems, choosing an appropriate decision interval becomes a critical challenge. Although reducing the decision interval can enhance control performance, it also increases the total number of decision steps required to reach the desired state from the initial state, thereby prolonging the policy training process. This creates a dilemma between training efficiency and policy performance, making it difficult for traditional DRL algorithms to effectively address high-dynamic control problems.

To improve the training efficiency of traditional DRL algorithms, we propose a novel event-triggered deep reinforcement learning (ETDRL) method in this work. Unlike the conventional time-triggered interaction mechanism of DRL algorithms, the ETDRL approach requires the agent to update its action only when specific events occur and maintain the same action between two consecutive events. In this way, the total number of decision steps is significantly reduced with the decision interval being increased, thereby accelerating the policy training process. Furthermore, we apply the developed ETDRL method to address the three-dimensional intercept guidance problem against maneuvering targets of speed advantage. The main contributions of this paper are summarized as follows:

1) An event-triggered Markov decision process (ETMDP) model is proposed to describe the ETDRL agent's event-based interaction with the environment. We theoretically demonstrate that ETMDP is a special class of MDPs with a temporally extended course of action persisting over an event rather than a time step. Leveraging the Lyapunov stability theory, we establish an event triggering condition for continuous control systems subjected to external disturbances, ensuring the system operates in a stable manner while minimizing unnecessary actions triggered by negligible changes in the environment.
2) To address the issue of control performance degradation when implementing a well-trained ETDRL policy in an event-triggered manner, we propose an event-triggered training and time-triggered execution (ETTE) framework for the ETDRL method. It allows the policy to be trained using event-triggered strategies and enables the well-trained policy to be executed in a time-triggered manner. The ETTE framework substantially improves training efficiency and maintains stable control performance, avoiding potential drawbacks caused by the event-triggered implementation.
3) An ETDRL guidance law is designed for intercepting maneuvering targets of speed advantage. Following the principle of nullifying the LOS angular rate, an ETMDP model tailored for the guidance problem is established. The event triggering condition is derived by analyzing the dynamics of the LOS angular rate, and a heuristic reward scheme is designed to simultaneously optimize energy consumption and convergence of the LOS angular rate. The proposed ETDRL guidance law demonstrates superior performance compared to traditional APN and DRL methods, as evidenced by its enhanced interception probability, enlarged effective capture region, and improved generalization capability.

The remainder of this paper is organized as follows. Section II presents the ETMDP model and the ETDRL algorithm. Section III develops the ETDRL guidance law. In Section IV, numerical simulations are conducted to validate the effectiveness of the proposed method. Conclusions are given in Section V.

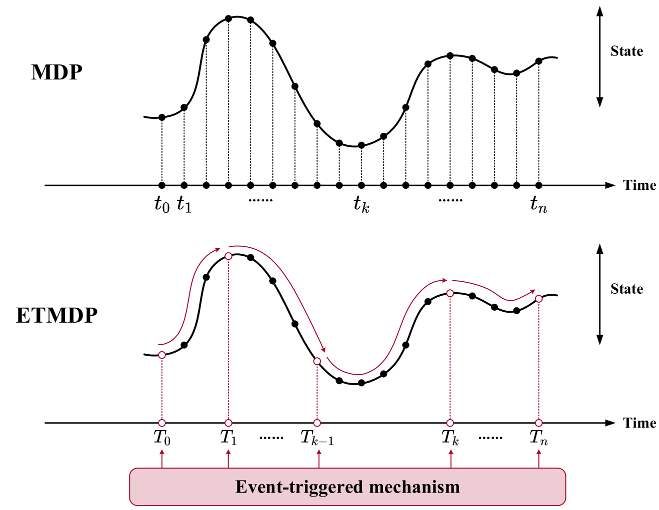

Fig. 1. MDP and ETMDP.

2. Event-triggered deep reinforcement learning

In this section, we provide a brief overview of the standard MDP model and extend it to an ETMDP by introducing the event-triggered mechanism. Then, an event triggering condition for nonlinear control systems subject to external disturbances is designed, and an event-triggered training and time-triggered execution framework is presented for the ETDRL algorithm.

2.1. Markov decision process

In the standard reinforcement learning framework, the environment is usually modeled as a MDP model and denoted as a five-tuple $(\mathscr{S},\mathscr{A},\mathscr{T},\mathcal{R},\gamma)$, where $\mathscr{S}$ is the state space, $\mathscr{A}$ is the action space, $\mathscr{T}:\mathscr{S}\times\mathscr{A}\times\mathscr{S}\rightarrow[0,1]$ is the state transition function, $\mathcal{R}:\mathscr{S}\times\mathscr{A}\times\mathscr{S}\rightarrow\mathbb{R}$ is the reward function and $\gamma\in(0,1]$ is the discount factor [39]. At each time step t, the agent interacts with the environment by observing a state $s_t\in\mathscr{S}$ and then choosing an action $a_t\sim\pi(s_t),\ \ a_t\in\mathscr{A}$ according to its policy π. After executing a_t, the environment state takes one-step transition to s_{t+1} according to $\mathscr{T}$ and a reward r_{t+1} is fed back to the agent. The objective of the agent is to find out an optimal policy mapping the observed states to actions by maximizing the expected cumulative reward:

$$\begin{aligned} V^{\pi}(s) &= \mathbb{E}\{r_{t+1}+\gamma r_{t+2}+\gamma^2 r_{t+3}+\cdots|s_t=s,\pi\} \\ &= \mathbb{E}\{r_{t+1}+\gamma V^{\pi}(s_{t+1})|s_t=s,\pi\} \\ &= \sum_{a\in\mathscr{A}}\pi(a|s)\left[r_s^a+\gamma\sum_{s'\in\mathscr{S}}p_{ss'}^a V^{\pi}(s')\right] \end{aligned} \tag{1}$$

where $\pi(a|s)$ is the probability of choosing action a in state s by using policy π, $p_{ss'}^a=\Pr\{s_{t+1}=s'|s_t=s,a_t=a\}$ is the one-step state transition probability and $r_s^a=\mathbb{E}\{r_{t+1}|s_t=s,a_t=a\}$ is the one-step expected reward.

The state-action value function is defined as

$$\begin{aligned} Q^{\pi}(s,a) &= \mathbb{E}\{r_{t+1}+\gamma r_{t+2}+\gamma^2 r_{t+3}+\cdots|s_t=s,a_t=a,\pi\} \\ &= r_s^a+\gamma\sum_{s'\in\mathscr{S}}p_{ss'}^a V^{\pi}(s') \\ &= r_s^a+\gamma\sum_{s'\in\mathscr{S}}p_{ss'}^a\sum_{a'\in\mathscr{A}}\pi(a'|s')Q^{\pi}(s',a') \end{aligned} \tag{2}$$

It represents the expected cumulative reward when taking action a in state s and henceforward following policy π. The optimal value function provides the optimal value of each state or state-action pair under the optimal policy:

$$\begin{aligned} V^*(s) &= \max_{\pi} V^{\pi}(s) \\ &= \max_{a\in\mathscr{A}}\mathbb{E}\{r_{t+1}+\gamma V^*(s_{t+1})|s_t=s,a_t=a\} \\ &= \max_{a\in\mathscr{A}}\left[r_s^a+\gamma\sum_{s'\in\mathscr{S}}p_{ss'}^a V^*(s')\right] \end{aligned} \tag{3}$$

$$\begin{aligned} Q^*(s,a) &= \max_{\pi} Q^{\pi}(s,a) \\ &= r_s^a+\gamma\sum_{s'\in\mathscr{S}}p_{ss'}^a\max_{a'\in\mathscr{A}}Q^*(s',a') \end{aligned} \tag{4}$$

2.2. Event-triggered Markov decision process

In essence, the standard MDP is formulated based on the time-triggered control. As shown in Fig. 1, the agent interacts with the environment at the minimum time scale $t_0,t_1,t_2,\cdots,t_n$ with a fixed decision time step Δt. To improve the learning efficiency of traditional DRL algorithms, we extend the MDP model to an event-triggered MDP by introducing the concept of the event-triggered control, where the control input is updated only after a specific event occurs and remains unchanged between two events.

As shown in Fig. 1, for ETMDP, the agent interacts with the environment over a larger time scale, where actions only need to be chosen at several discrete and irregular instants $T_0,T_1,T_2,\cdots,T_n$. Within the interval $[T_{k-1},T_k]$, the action $a_{T_{k-1}}$ is implemented at the minimum time scale using the zero-order hold strategy. The decision instant T_k is determined by the elaborate event triggering condition. For convenience, an event can be denoted as a triad $\varepsilon=(\mathscr{I},a,\chi)$ consisting of an initial state set $\mathscr{I}\subseteq\mathscr{S}$, an action $a\in\mathscr{A}$ and a terminal condition $\chi:\mathscr{S}\rightarrow\{0,1\}$. If the environment state is in the set $s\in\mathscr{I}$, an event is triggered, and then an action $a=\pi(s)$ is selected according to the agent's policy until the event terminates according to χ. The termination of the current event indicates a trigger of the next event, that is, when the current event terminates, it would turn to the next event automatically. When the environment terminates, the event terminates correspondingly.

Let $\varepsilon(s,a)$ denote an event which is initiated in state s at time t and an action a is chosen using policy π, the one-event expected reward can be defined as

$$r_s^{\varepsilon}=\mathbb{E}\{r_{t+1}+\gamma r_{t+2}+\cdots+\gamma^{k-1}r_{t+k}|\varepsilon(s,a)\} \tag{5}$$

where $t+k$ is the time when next event occurs. The one-event state transition probability is given as

$$p_{ss'}^{\varepsilon}=\sum_{k=1}^{\infty}\gamma^k p(s',k) \tag{6}$$

where $p(s',k)$ is the probability that the event terminates in $s'\in\mathscr{S}$ after k time steps. Then, the state value function and the state-action value function for ETMDP can be expressed as (7) and (8), respectively.

$$\begin{aligned} V^{\pi}(s) &= \mathbb{E}\{r_{t+1}+\gamma r_{t+2}+\cdots+\gamma^{k-1}r_{t+k}+\gamma^k V^{\pi}(s_{t+k})|\varepsilon(s,a)\} \\ &= \sum_{a\in\mathscr{A}}\pi(a|s)\left[r_s^{\varepsilon}+\sum_{s'\in\mathscr{S}}p_{ss'}^{\varepsilon}V^{\pi}(s')\right] \end{aligned} \tag{7}$$

$$\begin{aligned} Q^{\pi}(s,a) &= \mathbb{E}\{r_{t+1}+\gamma r_{t+2}+\cdots+\gamma^{k-1}r_{t+k}+\gamma^k V^{\pi}(s_{t+k})|\varepsilon(s,a)\} \\ &= \mathbb{E}\left\{r_{t+1}+\gamma r_{t+2}+\cdots+\gamma^{k-1}r_{t+k}+\gamma^k\sum_{a'\in\mathscr{A}}\pi(a'|s')Q^{\pi}(s',a')|\varepsilon(s,a)\right\} \\ &= r_s^{\varepsilon}+\sum_{s'\in\mathscr{S}}p_{ss'}^{\varepsilon}\sum_{a'\in\mathscr{A}}\pi(a'|s')Q^{\pi}(s',a') \end{aligned} \tag{8}$$

Accordingly, optimal value functions for the optimal policy of ETMDP are given as

Table 1
Event-triggered policy training algorithm for ETDRL.

Algorithm 1 Event-triggered Policy Training

Initialize parameters.
for $episode = 1, 2, \cdots, N$ **do**
  Initialize environment state s_0.
  Set $T_0 = 0$ and select action $a_{T_0} = \pi(s_0) + \mathrm{N}$.
  for $t = 0, 1, \cdots, n$ **do**
    Execute a_{T_k}, and then get the reward r_t and the next state s_{t+1}.
    if s_{t+1} satisfies event triggering condition (19) **then**
      Next event is triggered and $T_{k+1} = t + 1$.
      Store transition $\left(s_{T_k}, a_{T_k}, \sum_{t=T_k}^{T_{k+1}} \gamma^{t-T_k} r_t, s_{T_{k+1}}\right)$ into experience buffer.
      Select next action $a_{T_{k+1}} = \pi(s_{t+1}) + \mathrm{N}$.
      Random sample from buffer and update network parameters.
    end
    if environment terminates **then**
      Break
    end
  end for
end for

$$
\begin{aligned}
V^*(s) &= \max_{\pi} V^{\pi}(s) \\
&= \max_{a \in \mathscr{A}} \mathbb{E}\{r_{t+1} + \gamma r_{t+2} + \cdots + \gamma^{k-1} r_{t+k} + \gamma^k V^*(s') | \varepsilon(s,a)\} \\
&= \max_{a \in \mathscr{A}} \left[r_s^{\varepsilon} + \sum_{s' \in \mathscr{S}} p_{ss'}^{\varepsilon} V^*(s') \right]
\end{aligned} \tag{9}
$$

$$
\begin{aligned}
Q^*(s,a) &= \max_{\pi} Q^{\pi}(s,a) \\
&= \mathbb{E}\{r_{t+1} + \gamma r_{t+2} + \cdots + \gamma^{k-1} r_{t+k} + \gamma^k V^*(s_{t+k}) | \varepsilon(s,a)\} \\
&= \mathbb{E}\left\{r_{t+1} + \gamma r_{t+2} + \cdots + \gamma^{k-1} r_{t+k} + \gamma^k \max_{a' \in \mathscr{O}} Q^*(s_{t+k}, a') | \varepsilon(s,a)\right\} \\
&= r_s^{\varepsilon} + \sum_{s' \in \mathscr{S}} p_{ss'}^{\varepsilon} \max_{a' \in \mathscr{A}} Q^*(s', a')
\end{aligned} \tag{10}
$$

Remark 1. It is noted that the definitions of value functions take forms similar to those in standard MDPs, where the one-step expected reward and state transition probability are replaced by the one-event expected reward and state transition probability, respectively. It can be concluded that ETMDP is a special kind of MDP with a temporally-extended course of action persisting over an event rather than a time step. As a result, algorithms for MDP can be directly extended to the ETMDP case, such as, Q-learning updates the approximated Q-value function following $Q(s, a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a' \in \mathscr{A}} Q(s',a') - Q(s,a)]$, and the ETMDP version of Q-learning updates after each event terminates by $Q(s,a) \leftarrow Q(s,a) + \alpha\left[r_s^{\varepsilon} + \gamma^k \max_{a' \in \mathscr{A}} Q(s',a') - Q(s,a)\right]$, where α is the learning rate. The approximated Q-value function will converge to $Q^*(s,a)$ under similar conditions for the vanilla Q-learning [20].

2.3. Event-triggered deep reinforcement learning

(1) Event triggering condition

Some notations are given first. A continuous function $\alpha : [0,a) \rightarrow \mathbb{R}_0^+$, $a > 0$ is said to be of class $\mathscr{K}$ if it is strictly increasing and $\alpha(0) = 0$; it is said to be of class $\mathscr{K}_\infty$ if it is a class $\mathscr{K}$ function and satisfies $\alpha(z) \rightarrow \infty$ as $z \rightarrow \infty$.

Consider the following continuous control system subject to the external disturbance [40]:

$$
\begin{aligned}
&\dot{x}(t) = f(x(t), u(t), d(t)) \\
&u(t) = v(x(T_i)) \quad t \in [T_i, T_{i+1}), \ \ i \in \mathbb{S} \subseteq \mathbb{Z}_+
\end{aligned} \tag{11}
$$

where $x \in \mathbb{R}^n$ is the system state, $u \in \mathbb{R}^m$ is the control input, $d \in \mathbb{R}^{n_d}$ is the external disturbance, and $f : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^{n_d} \rightarrow \mathbb{R}^n$ denotes the system dynamics. The feedback control law $v : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is based on an event-triggered mechanism. The time sequence $\{T_i\}_{i \in \mathbb{S}}$ is determined by a pre-designed event-triggered mechanism, where $\mathbb{S} = \{0, 1, 2, \cdots\} \subseteq \mathbb{Z}_+$ is the set of the indices of the sampling instants and $T_0 = 0$.

Define the measurement error resulted from data sampling as

$$
e(t) = x(T_i) - x(t) \quad t \in [T_i, T_{i+1}), \ \ i \in \mathbb{S} \tag{12}
$$

and the controller can be rewritten as

$$
u(t) = v(x(t) + e(t)) \tag{13}
$$

Substituting (13) into (11) yields

$$
\begin{aligned}
\dot{x}(t) &= f(x(t), v(x(t) + e(t)), d(t)) \\
&= : g(x(t), e(t), d(t))
\end{aligned} \tag{14}
$$

Assumption 1. : System (14) is input-to-state stability (ISS) with e and d as the inputs if there exists a smooth Lyapunov function $V : \mathbb{R}^{n_x} \rightarrow \mathbb{R}$

Table 2
Time-triggered policy execution algorithm for ETDRL.

Algorithm 2 Time-triggered Policy Execution

Load the well-trained policy network
Initialize environment state s_0
for $t = 0, 1, \cdots, n$ **do**
  Select action $a_t = \pi(s_t)$
  Take action and observe next state s_{t+1}
  if environment terminates
    Break
  end
end for

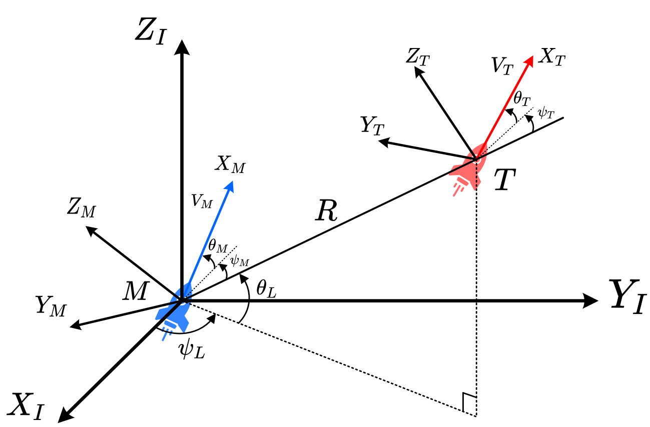

Fig. 2. Three-dimensional missile-target engagement.

satisfying

$$\underline{\alpha}_V(|x|) \leq V(x) \leq \overline{\alpha}_V(|x|) \tag{15}$$

$$\frac{\partial V}{\partial x} g(x(t), e(t), d(t)) \leq -\alpha(V(x)) + \max\{\gamma_1(|e|), \gamma_2(|\delta|)\} \tag{16}$$

where $\underline{\alpha}_V, \overline{\alpha}_V, \alpha, \gamma \in \mathscr{K}_\infty$.

Under Assumption 1, if the measurement error and external disturbance is restricted by

$$\sigma\alpha(V(x)) \geq \max\{\gamma_1(|e|), \gamma_2(|\delta|)\} \tag{17}$$

where $\sigma \in [0, 1]$ is a constant, then the derivate of Lyapunov function satisfies

$$\frac{\partial V}{\partial x} g(x(t), e(t), d(t)) \leq -(1-\sigma)\alpha(V(x)) \leq 0 \tag{18}$$

which guarantees a decreasing Lyapunov function. As a result, the triggering condition can be designed as

$$T_{k+1} = \inf\{t > T_k : \sigma\alpha(V(x)) \leq \max\{\gamma_1(|e|), \gamma_2(|\delta|)\}\} \tag{19}$$

Remark 2. The basic principle behind the triggering condition (19) is to attenuate the influence of measurement error and external disturbance by ensuring these quantities are smaller than a threshold related to the Lyapunov function. The inequality (17) is held to maintain system stability, and once the system dynamics violate the triggering condition, the stability will not be guaranteed if persisting in the current control input. Thus, the control input needs to be updated. By incorporating the triggering condition (19), the ETDRL agent is instructed to update its action when it is necessary to maintain the ISS property of the system. This mechanism allows for more efficient DRL policy training, as it effectively decomposes complex continuous control problems into several manageable sub-problems over an extended temporal horizon.

(2) Event-triggered Training and Time-triggered Execution Framework

The proposed ETMDP model establishes a novel interaction mechanism between the learning agent and the environment. However, the triggering condition (19) includes the unknown external disturbance, which cannot be obtained precisely when implementing the well-trained policy. In addition, implementing control strategies in an event-triggered manner usually leads to control performance degradation. To overcome these limitations, we propose a novel event-triggered training and time-triggered execution framework tailored to the ETDRL algorithm. The pseudo codes for policy training and execution are exhibited in Table 1 and Table 2, respectively.

In the policy training phase, as shown in Table 1, the network parameters and other algorithm hyper-parameters are initialized at the beginning. Then, for each interaction episode, the agent selects an action only when the environment state satisfies the predefined event-triggering condition, otherwise executing it until the next event is triggered. Events are automatically triggered at both the initial and terminal instants. Upon event termination, the transition tuple $\left(s_{T_k}, a_{T_k}, \sum_{t=T_k}^{T_{k+1}} \gamma^{t-T_k} r_t, s_{T_{k+1}}\right)$ is stored in a replay buffer, from which a mini-batch of transitions is randomly sampled to update the network parameters.

It is noted that the experience samples collected in the training stage do not cover the whole state space, but are limited to states meeting the event triggering condition. Nevertheless, due to the neural network's strong approximation ability to any nonlinear function and generalization ability, the well-trained policy network can infer appropriate actions when taking those states never met in the training set as inputs. For this reason, under the proposed ETTE framework, the well-trained policy is executed in a time-triggered way, that is, the agent's action is updated periodically with a fixed decision interval (see Table 2). In this way, it does not require additional information about the external disturbance to compute the triggering condition and the control performance can be guaranteed.

3. Guidance law based on ETDRL

In this section, the three-dimensional intercept guidance problem

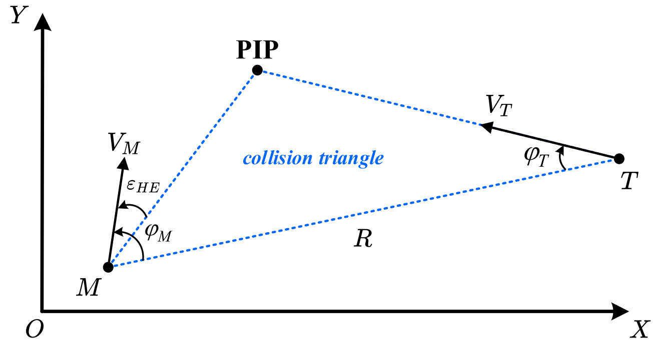

Fig. 3. Initial engagement geometry.

against a maneuvering target is investigated by using the proposed ETDRL algorithm. Firstly, the three-dimensional missile-target engagement with coupled dynamics is introduced. Secondly, by designing a heuristic immediate reward function and an event triggering condition, the original guidance problem is formulated as an ETMDP. Finally, an ETDRL guidance law optimized by the TD3 algorithm is developed.

3.1. Problem formulation

To simplify the dynamics of missile-target pursuit engagement, we make several assumptions: 1) the missile and target are point mass; 2) the angle-of-attack is small enough to be neglected; 3) the velocities of the missile and target are constant; 4) the head-on engagement is considered.

The three-dimensional missile-target engagement geometry is depicted in Fig. 2, where T and M denote the target and missile, respectively. $X_IY_IZ_I$ is an inertial reference frame. $X_TY_TZ_T$ and $X_MY_MZ_M$ are the velocity coordinate system of the target and missile, respectively. R denotes the missile-target relative distance, and the line-of-sight frame is defined with respect to the inertial frame by θ_L and ψ_L. The target and missile fly at the speed of V_T and V_M with their direction denoted as θ_T, ψ_T and θ_M, ψ_M, respectively, which are defined with respect to the LOS frame. Based on the geometric relationship, the three-dimensional missile-target engagement dynamics can be written as

$$\dot{R} = V_T\cos\theta_T\cos\psi_T - V_M\cos\theta_M\cos\psi_M \tag{20}$$

$$R\dot{\theta}_L = V_T\sin\theta_T - V_M\sin\theta_M \tag{21}$$

$$R\dot{\psi}_L\cos\theta_L = V_T\cos\theta_T\sin\psi_T - V_M\cos\theta_M\sin\psi_M \tag{22}$$

$$\dot{\theta}_M = \frac{a_{zM}}{V_M} - \dot{\psi}_L\sin\theta_L\sin\psi_M - \dot{\theta}_L\cos\psi_M \tag{23}$$

$$\dot{\psi}_M = \frac{a_{yM}}{V_M\cos\theta_M} + \dot{\psi}_L\tan\theta_M\cos\psi_M\sin\theta_L - \dot{\theta}_L\tan\theta_M\sin\psi_M - \dot{\psi}_L\cos\theta_L \tag{24}$$

$$\dot{\theta}_T = \frac{a_{zT}}{V_T} - \dot{\psi}_L\sin\theta_L\sin\psi_T - \dot{\theta}_L\cos\psi_T \tag{25}$$

$$\dot{\psi}_T = \frac{a_{yT}}{V_T\cos\theta_T} + \dot{\psi}_L\tan\theta_T\cos\psi_T\sin\theta_L - \dot{\theta}_L\tan\theta_T\sin\psi_T - \dot{\psi}_L\cos\theta_L \tag{26}$$

where a_{yT}, a_{zT} and a_{yM}, a_{zM} are the lateral accelerations of the target and missile, respectively. The missile autopilot dynamics is considered and modeled as a first-order lag system

$$\dot{a}_M = -\frac{1}{\tau}a_M + \frac{1}{\tau}a_M^C \tag{27}$$

where τ is the time constant and a_M^C is the acceleration command. The maneuverability of the target and missile is limited by a maximum value $a_T^{\max}$ and $a_M^{\max}$, respectively. In this paper, considering the hypersonic vehicle targets of speed advantage, it is assumed that the missile-to-target velocity ratio satisfies $V_M/V_T < 1$ and the maneuverability ratio satisfies $a_M^{\max}/a_T^{\max} < 2$.

Generally, the missile and target should be on a collision course to achieve interception. Specifically, as shown in Fig. 3, the velocity of the missile should point to the predicted intercept point (PIP) to form a collision triangle connecting M-PIP-T, and the following equality is satisfied

$$V_M\sin\varphi_M = V_T\sin\varphi_T \tag{28}$$

where φ_M and φ_T are defined as $\cos\varphi_M = \cos\theta_M\cos\psi_M$ and $\cos\varphi_T = \cos\theta_T\cos\psi_T$, respectively. Since there exists the handover error ε_{HE} between midcourse and terminal guidance stage, it is difficult to reach an ideal initial condition of collision triangle in practice. In this paper, an engagement scenario with a large initial heading error is considered. Additionally, the missile is presumed to be equipped with passive infrared seekers, which only provide bearings-only measurement.

Remark 3. The maneuvering target of speed advantage, the limited maneuverability of the missile, and large heading errors would significantly narrow the feasible solution domain and impose severe requirements on the guidance laws. Meanwhile, bearings-only measurement provides limited information to assist with decision-making. In this paper, the proposed ETDRL algorithm is utilized to solve this complicated guidance problem and provide an end-to-end guidance strategy, which takes the seeker's measurement as input and outputs the guidance command.

3.2. ETMDP modelling

Following the underlying principle of PN that nullifies the LOS angular rate to achieve a collision course, the ETMDP model for the three-dimensional guidance problem against maneuvering targets of speed advantage is designed.

Differentiating (21) yields

$$\dot{R}\dot{\theta}_L + R\ddot{\theta}_L = V_T\cos\theta_T\dot{\theta}_T - V_M\cos\theta_M\dot{\theta}_M \tag{29}$$

Substituting (20), (22), (23) and (25) into (29), it has

$$\ddot{\theta}_L = \frac{-2\dot{R}\dot{\theta}_L}{R} - \dot{\psi}_L\sin\theta_L\cos\theta_L - \frac{\cos\theta_M}{R}a_{zM} + \frac{\cos\theta_T}{R}a_{zT} \tag{30}$$

Differentiating (22) yields

$$\begin{aligned}&\dot{R}\dot{\psi}_L\cos\theta_L + R\ddot{\psi}_L\cos\theta_L - R\dot{\psi}_L\sin\theta_L\dot{\theta}_L = -V_T\sin\theta_T\sin\psi_T\dot{\theta}_T\\&+V_T\cos\theta_T\cos\psi_T\dot{\psi}_T + V_M\sin\theta_M\sin\psi_M\dot{\theta}_M - V_M\cos\theta_M\cos\psi_M\dot{\psi}_M\end{aligned} \tag{31}$$

Substituting (20), (21), and (23)-(26) into (31), it has

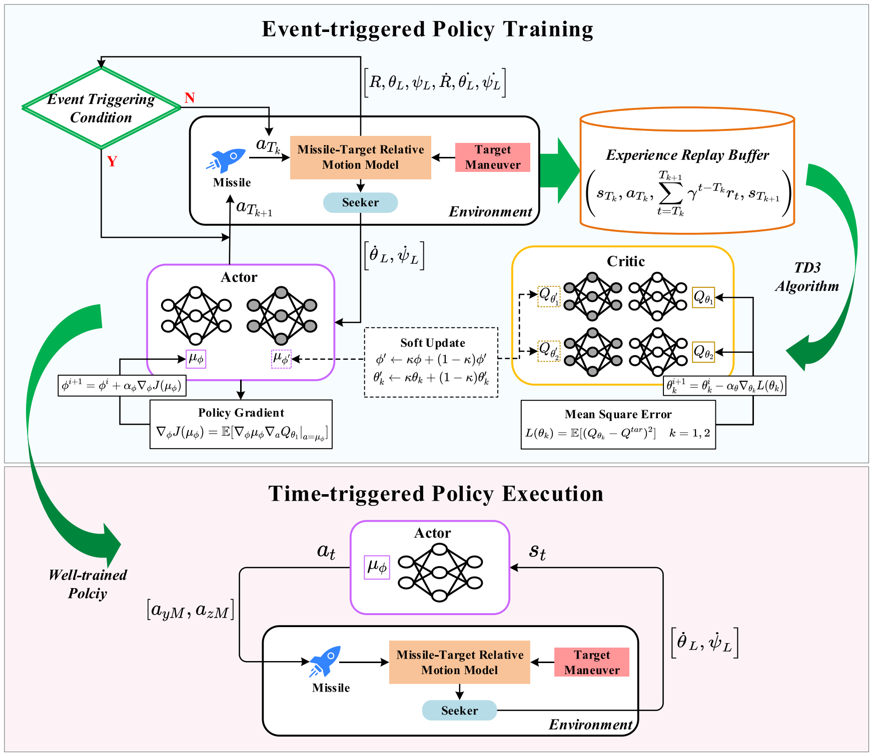

Fig. 4. ETTD3 guidance law.

$$\ddot{\psi}_L = -\frac{2\dot{R}\dot{\psi}_L}{R} + 2\dot{\theta}_L\dot{\psi}_L\tan\theta_L - \frac{\cos\psi_M}{R\cos\theta_L}a_{yM} + \frac{\sin\theta_M\sin\psi_M}{R\cos\theta_L}a_{zM} + \frac{\cos\psi_T}{R\cos\theta_L}a_{yT} - \frac{\sin\theta_T\sin\psi_T}{R\cos\theta_L}a_{zT} \tag{32}$$

Taking the LOS angular rates as new state variables, that is, $\boldsymbol{x} = \begin{bmatrix} \dot{\theta}_L \\ \dot{\psi}_L \end{bmatrix}$, the dynamics of LOS angular rates given by (30) and (32) can be rewritten as

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + \boldsymbol{B}\boldsymbol{u} + \boldsymbol{D} \tag{33}$$

where

$$f(\boldsymbol{x}) = \begin{bmatrix} -\dot{\psi}_L^2\sin\theta_L\cos\theta_L - \frac{2\dot{R}\dot{\theta}_L}{R} \\ 2\dot{\theta}_L\dot{\psi}_L\tan\theta_L - \frac{2\dot{R}\dot{\psi}_L}{R} \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} -\frac{\cos\theta_M}{R} & 0 \\ \frac{\sin\theta_M\sin\psi_M}{R\cos\theta_L} & -\frac{\cos\psi_M}{R\cos\theta_L} \end{bmatrix}, \boldsymbol{D} = \begin{bmatrix} \frac{\cos\theta_T}{R}a_{zT} \\ -\frac{\sin\theta_T\sin\psi_T}{R\cos\theta_L}a_{zT} + \frac{\cos\psi_T}{R\cos\theta_L}a_{yT} \end{bmatrix}$$

and $\boldsymbol{u} = \begin{bmatrix} a_{zM} \\ a_{yM} \end{bmatrix}$. The goal of guidance laws is to nullify the LOS angular rate, that is, $\boldsymbol{x} \to \boldsymbol{0}$, to reach a collision course.

With the definition of the Lyapunov function $V(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^{\mathrm{T}}\boldsymbol{x}$, by following (19), the event triggering condition for the three-dimensional guidance problem is designed as

$$T_{k+1} = \inf\{t > T_k : \sigma\alpha(V(\boldsymbol{x})) \le \max\{\gamma_1(|\boldsymbol{e}|), \gamma_2(|\boldsymbol{D}|)\}\} \tag{34}$$

where $\boldsymbol{e} = \boldsymbol{x}(T_i) - \boldsymbol{x}(t)$ is the measurement error, $\boldsymbol{D}$ is the external disturbance defined in (33), $\alpha(z) = 0.84z$, $\gamma_1(z) = 2.66z^2$, $\gamma_2(z) = 0.5z^2$ and $\sigma = 0.95$.

The ETDRL guidance agent generates the missile's acceleration command directly according to the measurement of onboard seeker. The agent's action is a vector consisting of the missile's lateral acceleration command

$$a = [a_{yM}, a_{zM}] \tag{35}$$

where each element is constrained within the range $[-a_M^{\max}, a_M^{\max}]$ and takes continuous value. As the missile is equipped with passive infrared seekers, only bearings information can be directly measured, and the environment state observed by the agent is given as

$$s = \left[\dot{\theta}_L, \dot{\psi}_L\right] \tag{36}$$

which only includes the missile-target LOS angular rates. Although the information of target's acceleration plays a key role in traditional guidance laws, it needs additional estimation, which brings not only estimation error but also online computational burden. For this reason, the raw sensory inputs are utilized in this work and mapped directly to the guidance commands with high-level useful information being extracted autonomously to assist decision-making.

The environment terminates when the sign of missile-target relative velocity turns positive and the terminal miss distance $R_f = R(t_f)$ is utilized to judge whether the target is intercepted or not, where t_f is the actual impact time. Accordingly, a terminal reward is designed to award the successful interception, which is given by

$$r(t_f) = \begin{cases} 50 + \lg(10/R_f) & R_f \le 10 \\ 0 & R_f > 10 \end{cases} \tag{37}$$

where a constant positive reward 50 is provided for the case that the terminal miss distance is less than 10 m, and a logarithmic function of the terminal miss distance is employed to inspire actions contributing to more precise interception. To improve the training efficiency and address the sparse feedback, an immediate reward function is designed by adopting the design principle of classic guidance laws about nullifying the LOS angular rate, and expressed as

$$r(t) = (1-\overline{R})\left(e^{-k\dot{\theta}_L^2} + e^{-k\dot{\psi}_L^2}\right) - \overline{R}\frac{a_{yM}^2 + a_{zM}^2}{(a_M^{\max})^2} \tag{38}$$

Table 3
Pseudo code for the ETTD3 guidance law.

Algorithm 3 ETTD3 guidance law

Initialize hyper-parameters.
Initialize actor and critic network parameters θ_1, θ_2, ϕ.
Initialize target networks $\theta_1' \leftarrow \theta_1,\ \theta_2' \leftarrow \theta_2,\ \phi' \leftarrow \phi$.
Initialize replay buffer B.
for $episode = 1, 2, \cdots, N$ **do**
  Initialize environment state s_0.
  Set $T_0 = 0$ and select action $a_{T_0} = \mu_\phi(s_0) + \mathrm{N}(0, \xi)$.
  for $t = 0, 1, \cdots, n$ **do**
    Execute a_{T_k}, and then get the reward r_t and the next state s_{t+1}.
    if s_{t+1} satisfies event triggering condition (34) **then**
      Next event is triggered and $T_{k+1} = t + 1$.
      Store transition $\left(s_{T_k}, a_{T_k}, \sum_{t=T_k}^{T_{k+1}} \gamma^{t-T_k} r_t, s_{T_{k+1}}\right)$ into B.
      Select next action $a_{T_{k+1}} = \mu_\phi(s_{t+1}) + \mathrm{N}(0, \xi)$.
      Sample a mini-batch of N_b transitions $\left(s_{T_k}, a_{T_k}, \sum_{t=T_k}^{T_{k+1}} \gamma^{t-T_k} r_t, s_{T_{k+1}}\right)$ from B
      Compute target Q value by adding policy noise:
      $\tilde{a} = \mu_{\phi'}\left(s_{T_{k+1}}\right) + \varepsilon, \varepsilon \sim \mathrm{clip}\left(\mathrm{N}(0, \sigma), -\varepsilon_{\max}, \varepsilon_{\max}\right)$
      $Q^{\mathrm{tar}} = \sum_{t=T_k}^{T_{k+1}} \gamma^{t-T_k} r_t + \gamma \min_{k=1,2} Q_{\theta_k'}\left(s_{T_{k+1}}, \tilde{a}\right)$
      Update critic networks: $\theta_i \leftarrow \arg\min_{\theta_i} \frac{1}{N_b} \sum \left(Q_{\theta_i}(s_{T_k}, a_{T_k}) - Q^{\mathrm{tar}}\right)^2, i = 1, 2$
      if t mod d **then**
        Update actor network by using policy gradient:
        $\nabla_\phi J(\mu_\phi) = \frac{1}{N_b} \sum \nabla_\phi \mu_\phi\left(s_{T_k}\right) \nabla_a Q_{\theta_1}\left(s_{T_k}, a\right)\Big|_{a=\mu_\phi\left(s_{T_k}\right)}$
      end if
      Soft update the target network parameters:
      $\phi' \leftarrow \kappa\phi + (1-\kappa)\phi',\ \theta_i' \leftarrow \kappa\theta_i + (1-\kappa)\theta_i'$
    end if
    if environment terminates **then**
      Break
    end if
  end for
end for

where $\overline{R} = \frac{R}{R_0}$, $R_0 = R(t_0)$ is the initial missile-target distance, and $k > 0$ is a positive constant. The time-varying weight coefficient $\overline{R}$ is utilized as a trade-off between control performance and energy consumption. The second term of Eq. (38) dominates and encourages less unnecessary maneuver to save energy when the missile-target relative distance is large, while the missile is demanded to spend the available maneuverability to pursue the target when getting much closer to the target.

3.3. Guidance law training

In this work, we adopt the TD3 algorithm [23] as a baseline to implement the ETDRL guidance law, namely event-triggered TD3 (ETTD3) guidance law. As shown in Fig. 4, there are an actor network $\mu_\phi(s)$, which outputs an action according to the observed state, and a pair of critic networks $Q_{\theta_1}(s, a)$, $Q_{\theta_2}(s, a)$, which evaluate the actions chosen by the agent. Target networks $\mu_{\phi'}, Q_{\theta_1'}, Q_{\theta_2'}$ with the same structure of the actor and critic networks are constructed to improve the

Table 4
Training Scenario Initialization Parameters.

Parameters (Unit)	Range	Parameters (Unit)	Range
R(km)	(40, 50)	θ_T(deg)	(10, 30)
θ_L(deg)	(30, 50)	ψ_T(deg)	(135, 225)
ψ_L(deg)	(70, 110)	ω(rad/s)	$\left(\frac{\pi}{5}, \frac{2\pi}{5}\right)$
ε_{HE}(deg)	(−10, 10)	R_{TM}(km)	(20, 40)
V_M(km/s)	2	V_T(km/s)	(2, 3)
$a_M^{\max}$(g)	8	$a_T^{\max}$(g)	5

Table 5
Hyper-parameters.

Hyper-parameter	Symbol	Value
Discount Factor	γ	0.99
Replay Buffer Size	$\lvert\mathscr{B}\rvert$	10^5
Batch Sampling Size	N_b	128
Actor Learning Rate	α_ϕ	10^{-4}
Critic Learning Rate	α_θ	10^{-4}
Exploration Noise	ξ	0.1
Policy Noise	σ	0.2
Clipped Noise Factor	$\varepsilon_{\max}$	0.5
Target Update rate	κ	5×10^{-3}
Actor Update Interval	d	2

stability of algorithm. In the policy training stage, the ETTD3 agent interacts with the environment and updates its action only when the event triggering condition (34) is satisfied. When an event is terminated, the experience sample $\left(s_{T_k}, a_{T_k}, \sum_{t=T_k}^{T_{k+1}} \gamma^{t-T_k} r_t, s_{T_{k+1}}\right)$ is collected into the replay buffer. At each time, a mini-batch of samples is randomly chosen from the replay buffer to update the network parameters. The well-trained ETTD3 guidance law is executed in a time-triggered way, that is, with a fixed decision interval. The pseudo code of the ETTD3 guidance law is summarized in Table 3 for clarity.

4. Simulation validation

In this section, numerical simulation and comparative analysis have been conducted to validate the effectiveness of the proposed ETDRL algorithm and ETTD3 guidance law. The simulation experiments were executed on a computer powered by 12th Gen Intel® Core (TM) i5-12400F CPU 2.50 GHz and NVIDIA GeForce RTX 3060. The missile-target engagement environment was developed using Python 3.7.10. The networks were built with Pytorch 1.13.0 and optimized through the Adam optimizer.

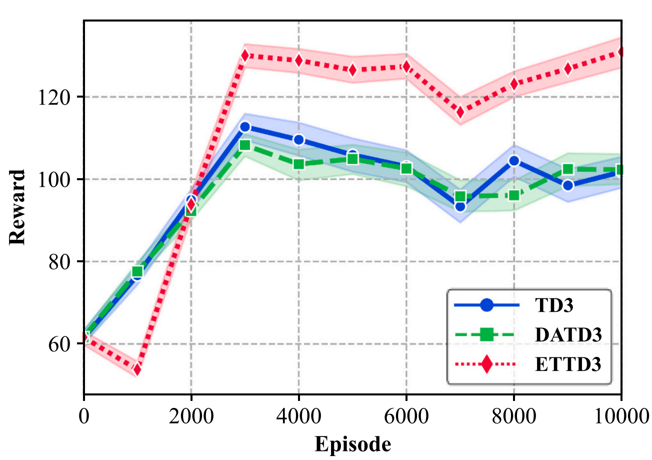

Fig. 5. Learning Curves.

Table 6
Runtime Comparison.

Method	Runtime (s)
TD3	6055.22
DATD3	9843.39
ETTD3	2796.46

4.1. Experimental settings

To reduce the randomness of the training process and enhance the adaptability of the guidance law, the training scenario is designed with a range of random initial conditions. As shown in Table 4, the initial environment states are uniformly sampled within their range. During the training phase, the target is programmed to perform the sinusoidal maneuver $a_T(t) = \begin{cases} a_T^{\max}\sin(\omega t) & R(t) \le R_{TM} \\ 0 & \text{otherwise} \end{cases}$, where the maximum acceleration $a_T^{\max}$ is set as 5 g, the maneuver frequency ω and the critical distance R_{TM} for starting maneuvering are uniformly distributed within their ranges. The missile's maneuverability $a_M^{\max}$ is 8 g, with the missile-to-target acceleration ratio $a_M^{\max}/a_T^{\max} = 1.6$. Furthermore, the time constant of autopilot dynamics is chosen as 0.1 s, and the environment is updated using the built-in ***solve_ivp*** function of the ***scipy*** toolkit with an integral time step of 0.1 s.

The hyper-parameters involved in the training procedure are listed in Table 5 and chosen identical to the original TD3 algorithm [23]. Both actor and critic networks consist of a three-layer fully connected structure, with hidden units of 32, 128 and 64 respectively. Each hidden layer is activated by the tanh function.

To validate the training performance of the proposed ETTD3 guidance law, we compare it with TD3 [23] and double actors TD3 (DATD3) [41] algorithms, where the DATD3 algorithm employs double actors to improve the sample efficiency compared to TD3. The hyper-parameters of TD3 and DATD3 algorithms are kept identical to those of ETTD3 for fair comparison. In addition, the ETMDP model designed in Sec. III-B, excluding the event triggering condition, is directly used for TD3 and DATD3 policy training.

4.2. Convergence performance analysis

The comparative results of learning curves for three methods, TD3, DATD3 and ETTD3, are shown in Fig. 5, where each algorithm is trained for 10,000 episodes and the episodic cumulative reward is averaged over 1000 episodes. At the first 3000 episodes, the learning curves show a consistently increasing trend for all three methods, and the ETTD3 method achieves significantly faster convergence compared to both TD3 and DATD3 algorithms. Then the learning curve goes steady hereafter, indicating that the learning process converges. It is obvious that the ETTD3 method achieves the highest average cumulative reward, suggesting that a higher interception probability and accuracy are achieved by ETTD3 compared to the other two methods. Table 6 provides a quantitative comparison of the computational efficiency among the three approaches. The baseline TD3 algorithm requires approximately 6055.22 s for training. In contrast, DATD3's additional actor network significantly increases its computational complexity, resulting in the longest runtime. By incorporating the event-triggered mechanism, ETTD3 achieves a substantial improvement in efficiency, completing the

Table 7
Interception Performance Comparison in Scenario 1.

Method	Miss distance (m)	Energy cost (m^2/s^3)
APN	6.32	62,736.70
TD3	2.31	26,198.85
DATD3	2.44	31,237.46
ETTD3	1.89	27,176.43

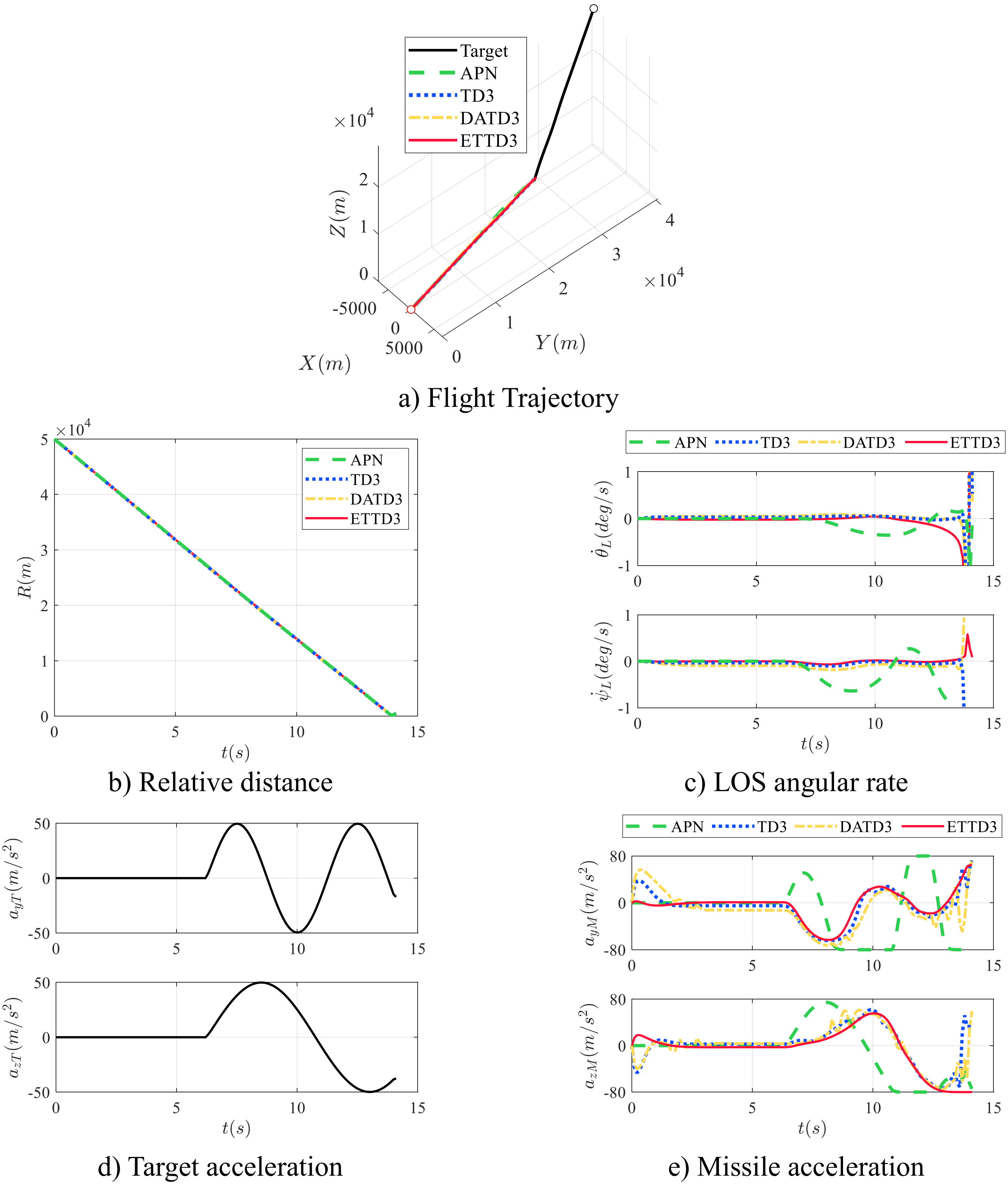

Fig. 6. Simulation Results in Scenario 1.

training process in just 2796.46 s, which is a nearly half reduction compared to TD3. These results strongly demonstrate that the proposed ETTD3 method exhibits superior learning efficiency and effectiveness when addressing complex guidance problems compared to both TD3 and DATD3 algorithms. These improvements attribute to the introduction of the event-triggered mechanism in the training process, which enables the agent to search for an optimal or suboptimal solution in an efficient way, thereby avoiding unnecessary exploration and enhancing training performance.

Table 8
Interception Performance Comparison in Scenario 2.

Method	Miss distance (m)	Energy cost (m^2/s^3)
APN	210.42	92,242.44
TD3	4357.05	34,123.28
DATD3	578.35	76,296.21
ETTD3	5.26	80,673.70

4.3. Guidance performance analysis

In this subsection, we test the guidance performance of TD3, DATD3 and ETTD3 guidance laws in various engagement scenarios. Additionally, the classic APN guidance law is employed for comparison, which is given as

$$a_{yM} = \left(1+\frac{N}{\cos\theta_M\cos\psi_M}\right)V_M\dot{\theta}_L\sin\theta_M\sin\psi_M$$
$$+\left(1+\frac{N}{\cos\theta_M\cos\psi_M}\right)V_M\dot{\psi}_L\cos\theta_M+\frac{\sin\theta_M\sin\psi_M}{\cos\theta_M\cos\psi_M}\widehat{a}_{zT}+\frac{1}{\cos\psi_M}\widehat{a}_{yT}$$
$$a_{zM} = \left(1+\frac{N}{\cos\theta_M\cos\psi_M}\right)V_M\dot{\theta}_L\cos\psi_M+\frac{1}{\cos\theta_M}\widehat{a}_{zT}$$

where the estimation of target acceleration $\widehat{a}_{yT}$ and $\widehat{a}_{zT}$ is introduced to compensate for the target's random maneuver, and in the simulation, a perfect target acceleration information is provided for APN.

(1) Scenario 1

The guidance laws are first tested in a scenario with the following initial parameters: , where the interceptor has the same velocity as the target and there is no initial heading error. Table 7 presents the terminal miss distances and energy costs of the four methods, where energy cost is defined as $\int_{t=0}^{t_f}\left(a_{yM}^2+a_{zM}^2\right)\mathrm{d}t$. It is evident that all four methods achieve successful interception with a terminal miss distance of less than 10 m, and the smallest miss distance of 1.89 m is obtained by the ETTD3 method. In addition, the energy consumption of TD3, DATD3 and ETTD3 is more than half lower than that of APN. The results show the superiority of DRL methods over traditional model-based approaches in both interception accuracy and energy consumption. Fig. 6 displays the flight trajectory of the missile and target, relative distance, LOS angular rate, target acceleration, and missile acceleration, respectively. In the first 6 s, as there is no initial heading error, the velocity vectors of the missile and target form a collision triangle, ensuring successful interception if the target does not maneuver. The target starts to maneuver at about 6 s, and the missile adjusts its acceleration accordingly to nullify the LOS angular rate and maintain the collision course. From Fig. 6(c), the three DRL methods show stronger capacity in controlling the LOS angular rate towards zero compared to APN. As shown in Fig. 6(e), the missile's acceleration command of APN increases rapidly and reaches saturation once the target begins to maneuver. In contrast, DRL methods only reach overload saturation when the missile and target are extremely close, indicating more efficient energy utilization. Moreover, the ETTD3 method demonstrates smoother guidance commands with less chattering during the final approaching phase compared to TD3 and DATD3.

(2) Scenario 2

The scenario 2 is extended with the target's velocity increasing to $V_T = 3\text{km/s}$ and the initial heading error setting as $\varepsilon_{HE} = -10^\circ$. All other parameters remain unchanged compared to scenario 1. As shown in Table 8, only the proposed ETTD3 method achieves a successful interception with a terminal miss distance of 5.26 m. The energy cost of ETTD3 is higher than the other two DRL methods but lower than APN. In this scenario, several challenges arise compared to scenario 1. As depicted in Fig. 7(b)-(c), the actual impact time reduces to about 12 s due to the increasing target speed. Additionally, the initial LOS angular rate is no longer zero because of a large initial heading error. These issues impose stricter requirements on the guidance law since the missile must exert greater control effort to correct the heading error and has less time to respond to the target's maneuver. As illustrated in Fig. 7(c)-(e), APN, DATD3 and ETTD3 methods initially attempt to correct the heading error by nullifying the LOS angular rate. However, the TD3 method has not learned a feasible solution for interception under these challenging conditions, and its LOS angular rate gradually diverges, resulting in a large terminal miss distance. When the target starts maneuvering, the guidance commands of both APN and DATD3 rapidly reach overload saturation, and their inability to effectively counteract the target's maneuvers ultimately results in failed interceptions. In contrast, despite the increasing target speed and large initial heading error, the proposed ETTD3 guidance law effectively utilizes the missile's limited maneuverability to achieve interception, demonstrating superior performance in handling this scenario.

(3) Monte Carlo Test

To enhance the statistical reliability of the results, 1000 Monte Carlo (MC) simulations are conducted under conditions with initial parameters randomly generated within the ranges specified in Table 4. The results of hit probability and average energy cost are shown in Table 9. The findings reveal that the proposed ETTD3 method achieves a hit probability of 89.9 %, demonstrating significant improvements compared to APN (78.4 % higher), TD3 (47.9 % higher), and DATD3 (40.1 % higher). Although the average energy cost of ETTD3 is slightly higher than that of the other three methods, the primary mission objective, to successfully capture the target, makes it logical to prioritize hit probability over energy efficiency.

(4) Capturability Analysis

In this subsection, the capturability of TD3, DATD3 and ETTD3 methods is analyzed. To evaluate their performance, simulation tests are conducted under extended scenarios that are unseen in the training process. Specifically, the target velocity ranges within $V_T \in [2,4](\text{km/s})$, and the initial heading error ranges within $\varepsilon_{HE} \in [-20,20](\text{deg})$. The results of capture region for each method are depicted in Fig. 8, where the red rectangle area with $V_T \in [2,3](\text{km/s})$ and $\varepsilon_{HE} \in [-10,10](\text{deg})$ represents the training scenario, while the area outside the red rectangle corresponds to the extended test scenarios. As illustrated in Fig. 8(a)-(b), both TD3 and DATD3 methods can only guarantee successful interception in the training scenario when the initial heading error is less than 4 deg. Furthermore, when the target's speed exceeds 3.5 km/s, the significant speed disparity between the target and missile makes intercept guidance challenging for these two methods. However, the proposed ETTD3 method not only achieves a hit probability of 100 % in the training scenario, but also shows capturability as V_T and $|\varepsilon_{HE}|$ increase. For instance, when the target's speed reaches 4 km/s, the ETTD3 method can still successfully intercept targets even with an initial heading error of less than 10 deg. Additionally, when the target and missile have equal speed, the capture region of ETTD3 is expanded to $|\varepsilon_{HE}| \le 16\text{deg}$. These results indicate that the proposed ETTD3 guidance law is of strong capacity to correct the collision course errors and handle maneuvering targets of speed advantage. Moreover, it outperforms TD3 and DATD3 methods by demonstrating robust generalization capability in unseen extended scenarios.

(5) Unseen Scenario Test

To further validate the proposed ETTD3 method's generalization capability for unseen scenarios, targets with a square wave maneuver are considered, which is given by $a_T(t) = \begin{cases} a_T^{\max}\text{sign}(\sin(\omega t)) & R(t) \le R_{TM} \\ 0 & \text{otherwise} \end{cases}$, where $\text{sign}(\cdot)$ is the symbolic function. 1000 Monte Carlo (MC) tests are conducted under the conditions specified in Table 4, and the results are listed in Table 10. The square wave maneuver imposes more stringent requirements on guidance laws because the acceleration command maintains its maximum value with the sign periodically reversed. Consequently, compared to the sinusoidal maneuver results presented in Table 9, all four methods exhibit a

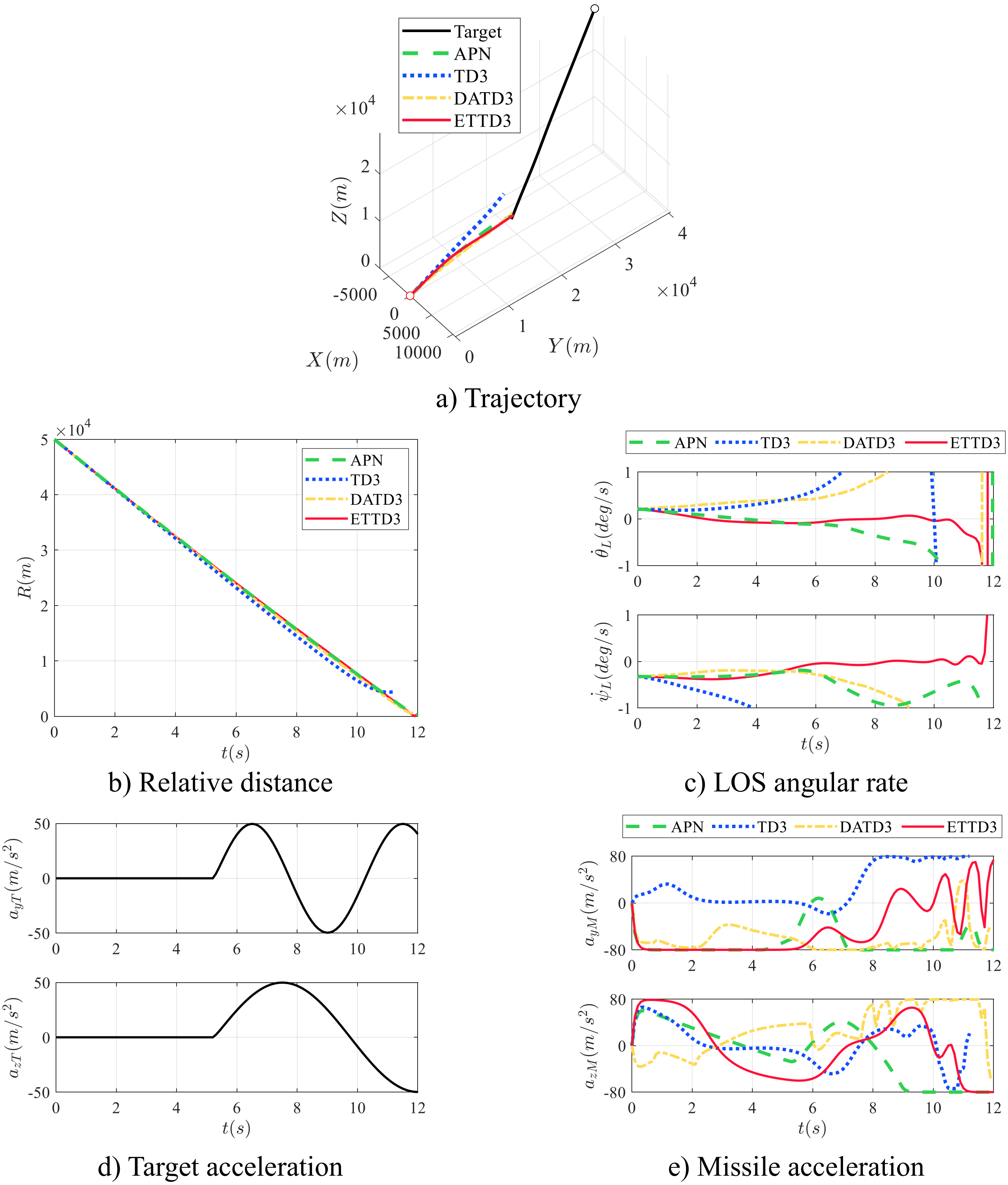

Fig. 7. Simulation Results in Scenario 2.

common trend of reduced hit probability: APN decreases by 5.7 %, TD3 by 29.8 %, DATD3 by 32.4 %, and ETTD3 by 11.8 %. The ETTD3 achieves the smallest decrease in hit probability among the three DRL methods. Furthermore, as evidenced by Table 10, the ETTD3 method still demonstrates a significant advantage in hit probability compared to APN (72.3 % higher), TD3 (65.9 % higher), and DATD3 (60.7 % higher). These results validate the strong generalization capabilities of the proposed ETTD3 method for handling unseen scenarios that deviate from the original sinusoidal maneuver patterns.

Table 9
Monte Carlo Simulation Results.

Method	Hit probability (%)	Average energy cost (m^2/s^3)
APN	11.5	47,313.54
TD3	42.0	42,530.78
DATD3	49.8	48,161.16
ETTD3	89.9	52,202.66

Table 10
Simulation Results in unseen scenarios.

Method	Hit probability (%)	Average energy cost (m^2/s^3)
APN	5.8	60,265.57
TD3	12.2	53,682.03
DATD3	17.4	65,649.51
ETTD3	78.1	77,405.17

4.4. Hyper-parameter sensitivity analysis

In this subsection, we perform a hyper-parameter sensitivity analysis to further investigate the effect of the proposed event-triggered training mechanism on policy convergence performance. Simulation experiments are conducted under the training scenario with initial parameters given in Table 4. The ETTD3 guidance law is trained by varying the value of σ in the designed event triggering condition (see Eq.(34)). The learning curves are plotted in Fig. 9. From Eq.(34), it can be intuitively observed that reducing σ increases the triggering frequency, while increasing σ makes the triggering condition more stringent. The results in Fig. 9 clearly indicate that the ETTD3 method achieves its best convergence performance when $\sigma = 0.95$, where not only does the convergence speed reach its peak but also the final converged reward is maximized. The results of 1000 Monte Carlo tests in the training scenario are shown in Table 11. The highest hit probability of 89.9 % is obtained when $\sigma = 0.95$. It shows a significant improvement compared to other settings: 16.5 %, 9.7 % and 20.2 % higher than the cases with $\sigma = 0.85$, $\sigma = 0.9$ and $\sigma = 1$, respectively. Based on these findings, it can be concluded that the choice of σ should strike a balance between being too large or too small to avoid compromising convergence performance. Additionally, although inappropriate selection of hyperparameters will hinder the improvement of learning efficiency for the ETTD3 method, its guidance performance remains superior compared to traditional APN and other DRL methods, as evident from Table 9 and Table 11.

5. Conclusion

This paper proposes a novel ETDRL algorithm, consisting of the ETMDP model and ETTE framework, to address inherently training inefficiency issues encountered by traditional DRL algorithms when solving high-dynamic continuous control problems. Based on the proposed method, we develop an ETTD3 guidance law for intercepting maneuvering targets of speed advantage with limited maneuverability. Numerical simulation results demonstrate significant improvements in policy training efficiency compared to traditional DRL methods when applied to complex guidance problems. The ETTD3 guidance law

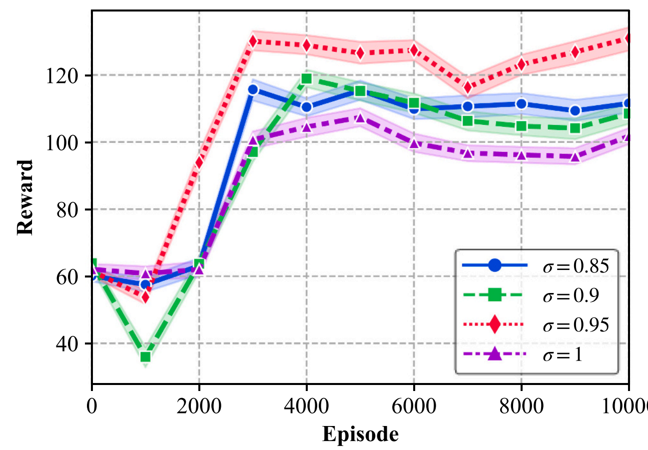

Fig. 9. Learning Curves for ETTD3 method with different σ.

Table 11
Monte Carlo Simulation Results for ETTD3 method with different σ.

σ	Hit probability (%)	Average energy cost (m^2/s^3)
0.85	73.4	47,071.24
0.9	80.2	49,204.39
0.95	89.9	52,202.66
1	69.7	47,900.14

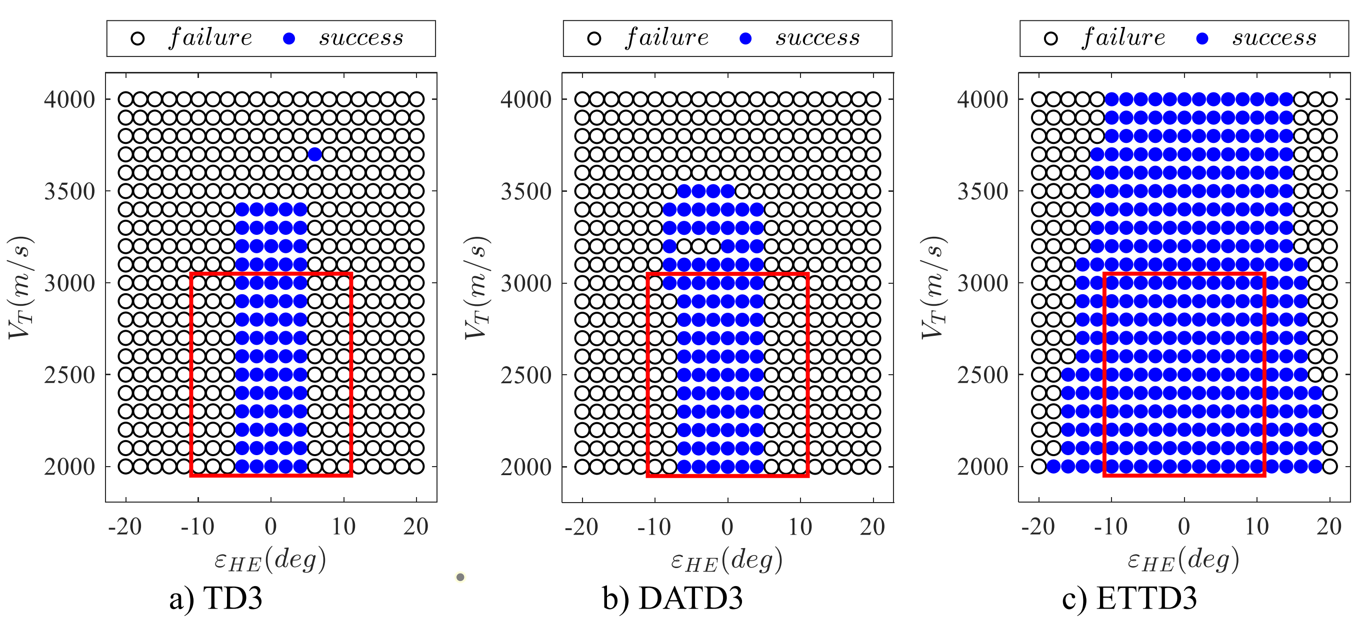

Fig. 8. Capture Region of TD3, DATD3 and ETTD3 methods.

exhibits superior performance over the classic APN guidance law and other DRL methods, as evidenced by its enhanced interception probability, enlarged effective capture region, and improved generalization capability. Additionally, a hyper-parameter sensitivity analysis is conducted to investigate how the event-triggered training mechanism affects policy convergence performance. Future research should extend these studies to more diverse and realistic engagement scenarios, such as those incorporating aerodynamics and measurement noise, to further enhance the algorithm's robustness and practicality in real-world applications. Moreover, consideration should be given to the deployment of the proposed algorithm on unmanned aerial vehicles for validation and testing.

CRediT authorship contribution statement

Wang Xu: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation. **Deng Yifan:** Writing – review & editing, Investigation, Funding acquisition. **Cai Yuan-Li:** Writing – review & editing, Supervision. **Jiang Haonan:** Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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